

TOPICAL REVIEW

The olympic brain. Does corticospinal plasticity play a role in acquisition of skills required for high-performance sports?

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Non-invasive electrophysiological and imaging techniques have recently made investigation of the intact behaving human brain possible. One of the most intriguing new research areas that have developed through these new technical advances is an improved understanding of the plastic adaptive changes in neuronal circuitries underlying improved performance in relation to skill training. Expansion of the cortical representation or modulation of corticomotor excitability of specific muscles engaged in task performance is required for the acquisition of the skill. These changes at cortical level appear to be paralleled by changes in transmission in spinal neuronal circuitries, which regulate the contribution of sensory feedback mechanisms to the execution of the task. Such adaptive changes also appear to be essential for the consolidation of a memory of performance of motor tasks and thus for the lasting ability of performing highly skilled movements such as those required for Olympic sports.

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It looks easy – and it may feel easy – but the seemingly effortless coordination of muscles acting at different body segments and joints, which is necessary in any high-performance sport, is still far beyond anything that can be achieved by robotic engineering and requires much more computational power than what can be delivered by the most powerful computers available. The beauty of any gymnastic move or the overwhelming power of a javelin throw relies on the ability of the nervous system to ensure that the right muscles are activated to the proper extent at the right time, and in the right sequence. It is trivial to note that this has been predominantly achieved by repeated training day in and day out for years on end and that the mastering of any sport at a level qualifying for an Olympic medal is not achieved by reading all available literature on the subject, but rather by the dull and tedious repeated performance of the motor task involved until it is mastered to perfection.

Neuroscience has made major advances within the past 10–20 years in order to understand the mechanisms involved in this gradual mastering of motor skills with repeated performance. A major breakthrough in our mechanistic understanding of these processes has been

the introduction of non-invasive electrophysiological and imaging techniques, which have made investigations of the intact human brain possible. Such studies have revealed how different parts of the human nervous system change activity as new tasks are learned and an understanding of the way that networks in the brain build and optimize the motor programs that are responsible for coordination of muscle activity involved in complex motor learning is now emerging. In this short review we will briefly describe plastic changes in the corticospinal pathway, which is responsible for conveying the command for voluntary movements from the cortex to the spinal cord.

Plasticity in the motor cortex and corticospinal tract

Whenever we optimize the performance of a motor skill by repeatedly performing the task, there is initially an expansion in the volume of the primary motor cortex and/or an increase in motor cortical excitability of the cortical representation devoted to the muscles involved in that task in the primary motor cortex. This was demonstrated originally by Pascual-Leone *et al.*

(1995a) for learning of sequences of finger movements (piano playing) using transcranial magnetic stimulation (TMS) by which corticospinal projections to specific muscles may be activated in the intact behaving human. Expansions of the cortical representation/increases in corticomotor excitability of specific muscles has since then been demonstrated in relation to the acquisition of a number of different motor skills involving both the hand (Pascual-Leone *et al.* 1995a; Latash *et al.* 2003), arm (Jensen *et al.* 2005) and leg (Perez *et al.* 2004). Using functional magnetic resonance imaging, expansions of the

local cerebral blood flow have also been demonstrated in relation to practice of a complex sequence of finger movements (Karni *et al.* 1995).

It is difficult to draw any firm conclusions regarding underlying neuronal mechanisms based on observations of the behaviour of muscular responses evoked by TMS, but there seems to be several processes at play (see Fig. 1 for schematic examples of different forms of plasticity in healthy volunteers and patients with brain lesions).

Voluntary activation of a muscle seems to involve a rapid (within some minutes of training), but short-lived

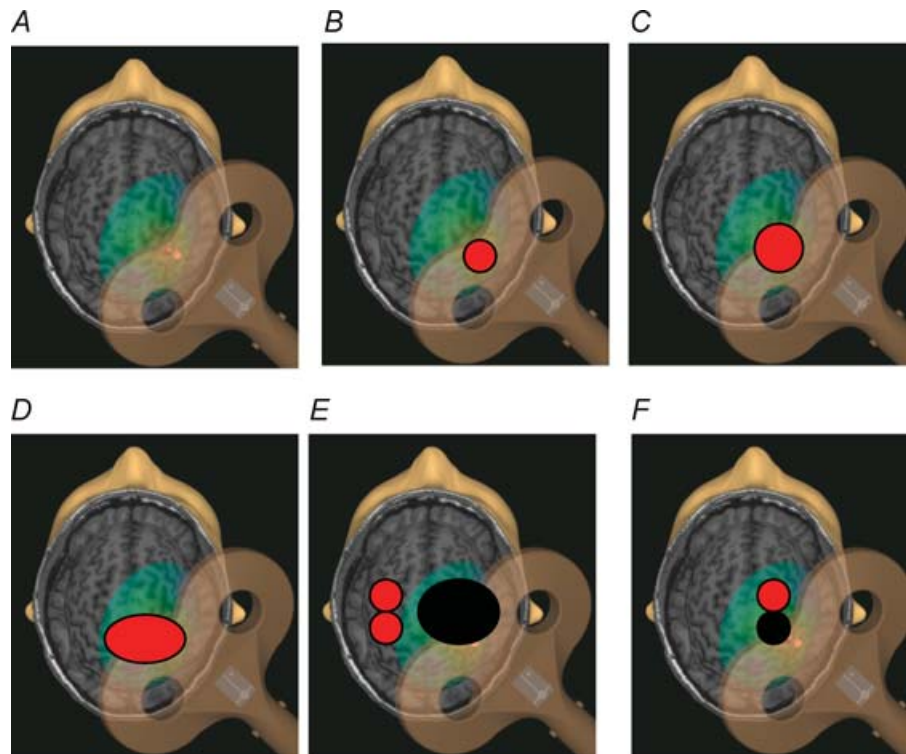


Figure 1. The purpose of this figure is to depict schematically some of the many different ways in which plasticity in the human motor cortex can be studied physiologically in health and disease

TMS application over the hand representation of M1 (Yousry *et al.* 1997) in a representative subject showing: A, maximum calculated induced field for this particular 8-shaped coil (light red dot) and position (green oval) using a stereotactic TMS device (Nexstim in this case) which localizes the target scalp position (in this case M1) on the same subject anatomical MRI; B, representative distribution of the motor map for a hand muscle (scalp locations which, upon TMS stimulation, elicited MEP responses from the target hand muscles) before training in a normal volunteer; C, centrifugal enlargement of the motor map (increase in the number of scalp positions which, upon stimulation, evoked MEP in target hand muscles) in a healthy volunteer, often described after motor training (see text). Note that a similar result in terms of centrifugal expansion of a motor map could be induced by an expansion in the motor cortical representation of that muscle or by an increase in motor cortical or spinal excitability of a topographically unchanged motor cortical representation (Ridding & Rothwell, 1995); D, enlargement in the motor map that is directionally more specific, showing a medial expansion of a hand muscle representation in M1 towards the upper arm representations, a more precise example of representational plasticity, as shown for example in amputees with phantom limb pain (Karl *et al.* 2001). For a detailed discussion of the differences between C and D see Ridding & Rothwell (1995), Ziemann *et al.* (1998, 2002); E, example of situations in which responses from a left hand or forearm muscles can be obtained by stimulation of the contralesional left M1 in cases of hemispherectomy (red posterior circle in the intact hemisphere) or even from the contralesional dorsal premotor cortex (red anterior circle) in patients with more severe forms of stroke (Benecke *et al.* 1991; Cohen *et al.* 1991; Johansen-Berg *et al.* 2002). The black oval depicts in diagrammatic form a large stroke or hemispherectomy engaging the right hemisphere; F, example of the involvement of ipsilesional dorsal premotor cortex, anterior to M1, in motor control of the paretic hand in patients with less severe stroke (Fridman *et al.* 2004).

(5–10 min) expansion of the representation or/and increase in corticomotor excitability of the muscle, which in all likelihood reflects a functional unmasking (possibly change in the level of GABAergic inhibition leading to increased excitability of corticospinal cells) of existing corticospinal projections to the muscle (Classen *et al.* 1998; Perez *et al.* 2004). In addition to changes in different muscle representations, repetition of stereotyped movements results in encoding of elementary motor memories in the primary motor cortex and probably other cortical areas that encode the kinematic details of the practiced movements, a form of memory for movement (Classen *et al.* 1998; Butefisch *et al.* 2000; Stefan *et al.* 2005). This may be regarded as a first step in skill acquisition as the expansion of the cortical representation is more pronounced the more engaging the task and the more learning required to accomplish it (Pascual-Leone *et al.* 1995; Perez *et al.* 2004; Jensen *et al.* 2005). One interesting feature of this form of memory formation for movement is that it can be encoded by action observation, that is, attentive observation of another individual performing a motor task can facilitate the representation of that movement in the primary motor cortex of the observer (Stefan *et al.* 2005). This form of action observation can enhance training effects under certain conditions (Celnik *et al.* 2006). One other relevant finding in the literature is that a variety of pharmacological agents can enhance this memory formation including dextroamphetamines (Butefisch *et al.* 2002) and dopaminergic agents (Floel *et al.* 2005a,b) in both healthy subjects and in patients with brain lesions like stroke. One consideration, very much under discussion these days, is the extent to which these pharmacological interventions influence performance in healthy volunteers, with obvious implications in relation to doping in sports.

Over the last few years, there has been an important effort to understand the mechanisms underlying reorganizational changes in the brain elicited by different forms of motor training. In one influential study, it was found that repetitive TMS applied to the primary motor cortex at a rate of 1 per second in the breaks inbetween training sessions may prevent the expansion of the cortical representation of the trained muscle and interfere with the consolidation of the learned task (Müllbacher *et al.* 2002). Repeated training of the task over several days or weeks leads to more or less long lasting expansion of the cortical representation (Karni *et al.* 1995; Pascual-Leone *et al.* 1995a, 1999; Jensen *et al.* 2005), which may reflect structural reorganization of the cortical networks activated by TMS or unmasking of cortical regions that before practice were engaged in performing other tasks (Sanes & Donoghue, 2000). Pascual-Leone *et al.* (1999) observed that the cortical representation of the trained muscle decreased again once the task had been acquired after some weeks of training despite continued

training. This indicates that this longer-lasting cortical expansion may also be more related to the acquisition process than to the improvement in the performance of the task per se. That the expansion of the cortical representation is not simply linked to muscle use, but to the way in which the muscle is used, is indicated by the studies by Carroll *et al.* (2002) and Jensen *et al.* (2005). Despite strength training every second day for several weeks leading to 20–30% increases in muscle strength, no changes in cortical excitability were observed in these studies. The picture emerging is consistent with the view that skill acquisition may recruit a large amount of cortical resources which, after the task is well learned, is performed by smaller cortical regions. If this is the case, the release of cortical resources after the task is learned would leave these areas available for engagement in new learning tasks. However, more work is required to test this general model.

Several studies have indicated that daily motor practice of certain tasks may lead to longer lasting and possibly permanent changes in the cortical representation of muscles. This was originally reported by Pascual-Leone *et al.* (1993) who observed that blind persons who used one hand to read Braille had a larger representation of the first dorsal interosseus muscle of that hand as compared with the other hand and that of blind control subjects. On the other hand, later investigations demonstrated that these changes in blind individuals reading Braille may fade after a weekend not practicing (Pascual-Leone *et al.* 1995b). Magnetic resonance imaging studies have also shown that musicians have a larger volume of the sensorimotor cortex than other subjects (Gaser & Schlaug, 2003), which may be related to a larger representation of the fingers in the sensory cortex (Elbert *et al.* 1995) and a higher excitability of the corticospinal projections to the fingers (Rosenkranz *et al.* 2007). Of direct relevance to sports, Pearce *et al.* (2000) have also reported that the cortical representation of the hand used for playing is larger in elite racquet players as compared with control subjects. It is also of interest that the acquisition of a motor skill as, for example, when learning to generate motor sequences, results not only in performance improvements in the practicing hand but also in the other resting hand, a process referred to as an intermanual transfer of skill learning. This process results in characteristic changes in intracortical inhibitory (predominantly GABAergic) mechanisms and also in interhemispheric (predominantly transcallosal) interactions (Perez *et al.* 2007).

Plasticity in spinal cord function associated with training

A very significant part of the corticospinal tract is indirectly connected to the spinal motoneurons via spinal

interneuronal networks and a large part of the descending motor command is thus influenced by changes in transmission in spinal networks (Nielsen, 2004). While it is widely accepted that cortical reorganization relates in different ways to the process of skill acquisition, it has so far been much less considered whether reorganization in the spinal networks might contribute as well. However, it has been known for quite some time that the monosynaptic spinal stretch reflex may be down- and up-regulated through operant conditioning and that the corticospinal tract is essential for this regulation (Wolpaw, 1994), but evidence for plastic changes related to motor learning have only appeared recently. The size of the soleus H-reflex, which is an equivalent of the stretch reflex evoked by electrical stimulation of the tibial nerve, is thus markedly smaller in ballet dancers than in other subjects with a similar level of daily physical activity (Nielsen *et al.* 1993). This may reflect habitually smaller reflexes in the dancers, but it may also be caused by lasting adaptive changes in the size of the reflex to the repeated performance of the specific motor repertoire of ballet dancers. One characteristic of ballet dancing is the necessity of high stability in the ankle joint which is ensured by co-contracting the ankle muscles. H-reflexes are indeed depressed during such co-contraction (Nielsen & Kagamihara, 1993) and training of co-contraction tasks lead to a progressive reduction in the reflex size (Monica Perez, Jesper Lundbye-Jensen & Jens Bo Nielsen; unpublished observations). The most likely mechanism for this depression is increased presynaptic inhibition of the Ia afferent terminals on the spinal motoneurons under the control of the corticospinal tract. Increased presynaptic inhibition has also been shown to be responsible for the depression of the H-reflex in relation to training of a task requiring integration of visual and proprioceptive feedback (Perez *et al.* 2005). Like the expansion of the cortical representation of muscles, this depression appears to be closely associated to the acquisition of the motor task, rather than its performance, and it may in this specific case relate to the necessity of integrating the feedback from the two sensory systems at a cortical level during the acquisition of the new skill. Plastic adaptive changes at spinal and cortical level thus appear to go hand in hand to ensure that the motor control system is optimally tuned for the particular motor task at hand (Perez *et al.* 2005; Meunier *et al.* 2007).

Implications of these findings in health and disease

This amazing ability of the brain to reorganize itself in response to new environmental challenges in the setting of training a new task or sport has elicited the interest of sport scientists as well as clinicians. The question that has emerged is that perhaps this ability could operate in the process of rehabilitation following injuries like

bone fractures, amputations or after brain lesions like stroke.

Cortical, and probably spinal plasticity as well, appear to play an important role in functional rehabilitation after brain lesions like stroke. For example, after a lesion in one side of the brain, a motor task that used to engage discreet areas of the motor system recruit a much larger network in order to perform the same or a diminished motion (Ward & Cohen, 2004; Butefisch *et al.* 2005; Ward *et al.* 2006). What has been an area of active investigation is determining what role each of these areas plays in generating these movements in the process of motor training. One reason this has elicited so much attention is because nowadays there are non-invasive tools that allow us to modulate excitability in different parts of the brain on purpose (Nudo *et al.* 1990). Recent animal studies have shown that direct epidural stimulation of the primary motor cortex surrounding a small infarct in the lesioned hemisphere elicits improvements in motor function (Brown *et al.* 2006). In humans, proof of principle studies from different laboratories have shown that non-invasive transcranial magnetic stimulation and direct current stimulation that increased excitability within the ipsi-lesional primary motor cortex (Gow *et al.* 2004; Khedr *et al.* 2005; Hummel *et al.* 2005; Hummel & Cohen, 2006; Talelli *et al.* 2007; Hesse *et al.* 2007) or decreased excitability in the contra-lesional primary motor cortex (Takeuchi *et al.* 2005; Fregni *et al.* 2005; Mansur *et al.* 2005; Boggio *et al.* 2006) results in improvement in motor function in patients with stroke. Possible mechanisms mediating these effects may include the correction of abnormally persistent interhemispheric inhibition when moving the paretic hand, a disorder correlated with the magnitude of impairment (Duque *et al.* 2005, 2007). Activation of sensory afferent inputs, especially in combination with stimulation of specific motor areas, also appears as an efficient means of promoting plastic changes at both cortical (Ridding & Rothwell, 1999; Khaslavskaja *et al.* 2002) and spinal level (Perez *et al.* 2003). Such interventions may with some likelihood be part of the treatment repertoire offered by future neurorehabilitation units for the benefit of disabled people and – who knows – may also be found in the gym of athletes training for the 2012 Olympics or alternatively on the list of illegitimate doping remedies. What is of main importance, and which is what we have tried to point out with this review, is that the development of non-invasive imaging and electrophysiological techniques have made objective measures available that allow us to directly monitor functionally relevant changes in neuronal circuitries in the brain and spinal cord induced by training or other interventions. This gives us the possibility of evaluating and guiding the interventions based on an unprecedented mechanistic understanding.

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